AI INTERVIEW COACH

Document 1 of 3

Project Planning and Management

Project Proposal, Project Plan, Task Assignment and Roles, Risk Assessment and Mitigation Plan, and Key Performance Indicators.

Item	Detail
Project	AI Interview Coach
Team	Team 147
Course / Track	AI Capstone Project – Hybrid (Deep Learning / NLP + LLMs & RAG)
Document set	Project Documentation Package (3 documents)
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Status	Consolidated draft for academic review

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About This Document

This document is one of three that consolidate the nine Team 147 source documents (Project Proposal and Plan, Task Assignment and Rules, Risk Assessment and Mitigation Plan, KPIs Framework, Feedback and Evaluation Specification, Requirements extract, Software Requirements Specification, Software Design Document, and Stakeholder Analysis / User Stories / Use Cases). The source documents are treated as the source of truth. Status labels used throughout:

Label	Meaning
Confirmed	Stated in the source documents as a decision, scope item or documented fact.
Proposed	Stated in the sources as a proposal, target or recommendation that the team has not yet confirmed.
Recommendation (review)	Added during this consolidation review; not present in any source document.
To Be Confirmed	Required information (for example a calendar date or an owner) is missing or conflicting.
Not Provided in the Source Documents	No evidence exists in the supplied files.

Scope of This Document

Document 1 covers sections 1.1 to 1.5. Document 2 (Lecturer Review) and Document 3 (Requirements Gathering) are referred to by section number rather than repeated. Unresolved conflicts between source documents are listed in Appendix A and are flagged in the relevant sections.

1.1 Project Proposal

1.1.1 Project Overview and Background

AI Interview Coach is a web platform that helps job seekers, students and recent graduates prepare for job interviews. A user uploads a CV, selects a target job role and answers interview questions. The system scores each answer, reports a confidence score, highlights strengths and weaknesses, and suggests concrete improvements. A generative AI layer produces personalized interview questions and ideal-answer suggestions based on the user's CV and the chosen role.

The project combines NLP-based answer evaluation (Sentence Transformers, BERT / DistilBERT) with LLM-based generation (OpenAI API / Amazon Bedrock) and retrieval over curated interview and career resources (RAG). It is delivered through a React.js interface and a FastAPI backend deployed on AWS. The project belongs to the Hybrid track (Deep Learning / NLP + LLMs & RAG) of the AI Capstone Project.

1.1.2 Problem Statement and Motivation

Interview performance strongly affects hiring outcomes, yet most candidates prepare without realistic practice or objective feedback. The common alternatives each fall short:

- **Static question banks and prep guides** are generic and do not reflect the candidate's own experience or the target role.
- **Human mock interviews** (mentors, career centers) are valuable but scarce, hard to schedule and difficult to repeat often.
- **Self-assessment** is subjective, so candidates often cannot tell which answers are weak or why.

The value proposition is personalized, CV-aware, immediate feedback with an objective AI-based score, giving targeted coaching instead of one-size-fits-all tips.

1.1.3 Objectives and Expected Outcomes

Objective numbering follows the Project Proposal (O1–O6). The Task Assignment document uses a different five-objective numbering; its refinements are shown in the last column. Targets in that column are **Proposed** unless stated otherwise (see 1.5 and Appendix A, C-06).

ID	Objective	Measure of success (Proposal)	Refinements in other sources
O1	Build an end-to-end working platform (CV upload → interview session → feedback dashboard).	All core user flows work in a deployed environment.	Task Assignment: flow completes in a deployed environment with no manual intervention.
O2	Develop an answer-evaluation model that scores answers by semantic similarity to ideal responses.	Strong positive correlation with human-rated benchmark scores (target set during evaluation, e.g. Spearman ≥ 0.6).	Task Assignment: $\rho \geq 0.60$, to be confirmed after Sprint 2 baseline. KPI Framework: Pearson $r \geq 0.75$ and Spearman $\rho \geq 0.72$.
O3	Generate personalized questions and ideal answers from the CV and target role using an LLM.	Reviewers rate generated questions relevant in at least 80% of sampled cases.	Task Assignment: ≥ 90% "relevant" in a 50-sample review, plus schema and safety validation. KPI Framework: mean question quality ≥ 4.0 / 5.
O4	Deliver actionable feedback (strengths, weaknesses, suggestions) for every answer.	Feedback for 100% of submitted answers; average response time under about 10 seconds.	Task Assignment and KPI Framework: p95 evaluation latency ≤ 3 s.
O5	Deploy a scalable, monitored cloud solution on AWS.	Publicly reachable application with CloudWatch dashboards and alerts active.	Task Assignment adds CI/CD to staging, and CVs encrypted at rest and in transit.
O6	Deliver complete documentation and a final demonstration.	Final report, user guide and demo completed on schedule.	—

Expected outcomes: a deployed, monitored platform; a validated scoring pipeline with an evaluation report; a GenAI/RAG module; the project documentation package; and a final demonstration. The deliverables for each phase are listed in 1.2.5.

1.1.4 Project Scope

In scope	Out of scope (this project phase)
User-facing React.js interface: CV upload, role selection, interview session, results dashboard and session history.	Video-based analysis such as facial expressions or body language.
FastAPI backend: CV handling, question delivery, answer submission, results retrieval and authentication.	Native mobile applications.
CV parsing (PDF/DOCX to text) and job-description / role processing.	Employer-side tools such as candidate screening, ranking or ATS / job-board integration.
Answer evaluation with Sentence Transformers and BERT / DistilBERT (semantic similarity, confidence score, strengths / weaknesses).	Fine-tuning or training a large language model from scratch.
GenAI integration (OpenAI API or Amazon Bedrock) for personalized questions, ideal answers, follow-ups and suggestions, supported by RAG over curated resources.	Live human-recruiter matching or paid subscriptions.
Relational database (PostgreSQL / MySQL on AWS RDS) for users, sessions, answers and results.	Multi-language support beyond English.
AWS deployment, monitoring and CI/CD (EC2, S3, RDS, API Gateway, CloudWatch, GitHub Actions).	
Testing, model-quality evaluation and project documentation.	

Stretch goals (only if time allows): voice answers transcribed with Whisper and evaluated by the same pipeline; progress analytics across sessions and exportable feedback reports. The Task Assignment document enables voice answers only if the Sprint 2 benchmark passes.

> *Scope conflict (Appendix A, C-03): the Proposal and Task Assignment treat voice input as a stretch goal, whereas the SRS and Requirements extract include optional voice input as a Medium-priority requirement (Document 3, FR-035 to FR-038). Both views are preserved; the team must decide.*

1.1.5 Target Users and Stakeholders

The primary users are job seekers, university students and fresh graduates preparing for interviews across a variety of roles. The full stakeholder analysis is in Document 3, section 3.1.

1.1.6 Proposed Solution and Main Features

1. The user uploads a CV (PDF/DOCX) and selects a target job role.
2. The system generates role-specific, CV-aware interview questions.
3. The user answers each question in text (voice answers are optional; see scope conflict above).
4. An evaluation model compares the answer with ideal responses and produces a confidence score plus strengths and weaknesses.
5. The GenAI layer provides improvement suggestions, an ideal-answer example and optional follow-up questions.
6. Results are stored per session so the user can track progress on a dashboard.

Table 1.1 – Technical approach

Layer	Approach and technology
Frontend	React.js: authentication, CV upload, interview session screen, results dashboard.
Backend	Python with FastAPI: REST API, request validation, orchestration of AI modules.
Answer evaluation (NLP / DL)	Sentence Transformers for embeddings and cosine similarity; BERT / DistilBERT as classification / regression head or baseline comparison; confidence score and strengths / weaknesses derived from similarity and key-concept coverage. The composite scoring formula is specified in the Feedback and Evaluation Specification.
Generative AI (LLM + RAG)	OpenAI API or Amazon Bedrock for question generation, ideal answers and suggestions; retrieval over curated question banks and career resources to ground outputs.
Speech (optional / stretch)	Whisper for speech-to-text.
Data storage	PostgreSQL / MySQL on AWS RDS; AWS S3 for uploaded CV files. Vector store for RAG is undecided (Appendix A, C-05).
Infrastructure	AWS EC2, API Gateway, S3, RDS, CloudWatch; GitHub Actions for CI/CD.

System workflow: the user interacts with the React interface; the FastAPI backend validates and stores data in RDS; NLP models score the answer while the GenAI layer generates feedback, follow-ups and ideal answers; results return to the frontend while CloudWatch monitors health and performance.

Table 1.2 – Data plan

Data source	Format	Use
User CVs (testers during development)	PDF/DOCX converted to text	Personalization of questions and feedback
Job postings / descriptions (public APIs or scraping, respecting site terms)	Unstructured text	Role understanding, question generation
Interview question / answer banks and career resources (curated, public)	Text	RAG knowledge base and ideal-answer references
Synthetic Q&A pairs generated with an LLM and reviewed by the team	Text	Expanding training and evaluation coverage

The expected initial volume is roughly 10,000–50,000 text documents. Data will be cleaned, de-duplicated and anonymized where it contains personal information.

1.1.7 Assumptions, Constraints and Dependencies

Assumptions

- The project runs within one academic term. The Proposal assumes 14 weeks; the Task Assignment assumes 10 weeks. Both can be shifted to the course calendar (Appendix A, C-01).
- Answers are submitted as text in the core version.
- Cloud and API usage stays within the team's budget, free-tier limits or academic credits (budget value To Be Confirmed; the KPI Framework cites US$150).

Constraints

- Uploaded CVs contain personal data and must be handled with access control, encrypted storage, no sharing with third parties beyond the LLM provider's API, and deletion on request.
- The SRS design constraints fix AWS EC2, S3, RDS, API Gateway and CloudWatch as infrastructure, and Sentence Transformers / BERT / DistilBERT with Whisper as the AI components.

Dependencies

- External services: OpenAI API / Amazon Bedrock and AWS, used under the providers' terms; Bedrock model-access approval may take days (Risk R06).
- Open decisions that block design work: database engine, GenAI provider and failover order, vector store, voice scope and calendar dates (Appendix A).

1.2 Project Plan

1.2.1 Management Approach

The Proposal specifies an Agile, sprint-based approach: weekly team meeting, sprint planning and review with a demo of working increments, a shared task board (Trello, Jira or GitHub Projects), GitHub version control with feature branches and pull-request reviews, and a definition of done (code reviewed and merged, tested, documented and demonstrated). Scope changes are discussed at the weekly meeting and recorded on the task board; stretch goals are pulled in only after milestone M5.

1.2.2 Planning Baseline and Schedule Status

The source documents contain two different schedule baselines. Neither includes calendar dates; the date of the course calendar is To Be Confirmed (Task Assignment, open decision A5). The Proposal baseline is used for the phases, Gantt chart and milestones below because it is the only source with a week-by-week task schedule. The Task Assignment roadmap is shown in 1.2.6.

Source	Duration	Structure
Project Proposal and Plan	14 weeks (assumed)	Seven two-week sprints; seven phases; eight milestones (M1–M8).
Task Assignment and Rules (v1.0, 16 Oct 2026)	10 weeks (assumed)	Four sprints (Weeks 1–2, 3–5, 6–8, 9–10).
Software Design Document (v1.0, 16 Oct 2026)	About 7–9 weeks (sum of phase durations)	Six implementation phases of 1–2 weeks each.
KPI Framework (v1.0)	Not stated	Refers to “Sprints 5–6” for integration and staging, implying at least six sprints.

> *Progress status: no source document reports completed work or actual progress. All implementation items are therefore Planned. The documentation baseline (SRS, SDD, Task Assignment, Risk Plan, KPI Framework, Feedback Specification) exists at version 1.0 dated 16 October 2026.*

1.2.3 Project Phases

Phase	Weeks	Main activities
1. Planning and requirements	1–2	Finalize scope, user stories, architecture, repository and task board; approve proposal and plan.
2. Data and infrastructure foundation	2–5	Collect and clean data; design database schema; provision AWS (EC2, S3, RDS, API Gateway, CloudWatch).
3. Core development	3–10	Build backend API, frontend screens, answer-evaluation models and GenAI / RAG integration in parallel.
4. Integration	9–11	Connect frontend, backend, database and AI modules into a complete end-to-end flow.
5. Testing and evaluation	10–12	Functional, integration and performance testing; model evaluation; bug fixing.
6. Deployment and monitoring	11–13	Production deployment on AWS, monitoring dashboards and alerts, security review.
7. Documentation and delivery	12–14	Final report, user guide, demo preparation and presentation.

1.2.4 Gantt Chart

Shaded cells mark the weeks in which each task is active (Proposal baseline, 14 weeks; calendar dates To Be Confirmed).

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Requirements, design, and planning	■	■												
Data collection and preprocessing		■	■	■	■									
Database schema and AWS setup		■	■	■										
Backend API (FastAPI)			■	■	■	■	■	■						
Answer-evaluation models (NLP/DL)				■	■	■	■	■	■					
GenAI and RAG integration					■	■	■	■	■					
Frontend (React.js)				■	■	■	■	■	■	■				
System integration									■	■	■			
Testing and evaluation										■	■	■		
Deployment and monitoring											■	■	■	
Documentation, demo, and presentation												■	■	■

1.2.5 Milestones and Deliverables

Table 1.3 – Milestones (Proposal baseline)

ID	Week	Milestone	Completion criteria	Calendar date
M1	2	Project kickoff complete	Proposal and plan approved; repository, task board and architecture diagram ready.	To Be Confirmed
M2	4	Foundation ready	AWS environment and database schema live; first data pipeline working.	To Be Confirmed
M3	6	Core backend and UI prototype	CV upload, session and answer-submission endpoints working; clickable frontend prototype.	To Be Confirmed
M4	8	Alpha AI modules	Baseline evaluation model producing scores; GenAI question and ideal-answer generation working through the API.	To Be Confirmed
M5	10	MVP end-to-end	Full flow: CV upload → questions → answer → score and feedback → dashboard.	To Be Confirmed
M6	12	Testing complete	Test report and model evaluation results delivered; critical bugs fixed.	To Be Confirmed
M7	13	Production deployment	Application live on AWS with CloudWatch monitoring and alerts.	To Be Confirmed
M8	14	Final delivery	Final report, user guide, demo and presentation delivered.	To Be Confirmed

Table 1.4 – Deliverables by phase

Deliverable	Description	Owner	Due	Phase
Project proposal and plan	Proposal and plan with architecture overview	All members	Week 2	1
Database schema and scripts	Tables for users, sessions, questions, answers, results	Database Engineer	Week 4	2

Deliverable	Description	Owner	Due	Phase
AWS infrastructure	EC2, S3, RDS, API Gateway, CloudWatch configured	Cloud / DevOps Engineer	Week 4	2
Cleaned datasets and RAG knowledge base	Processed CVs / job postings, question banks, synthetic Q&A	ML Engineer, GenAI Integration Lead	Week 5	2
Backend API	FastAPI service with documented endpoints (OpenAPI)	Backend Developer	Week 8	3
Answer-evaluation model	Scoring pipeline with evaluation report	Machine Learning Engineer	Week 9	3
GenAI module	Question generation, ideal answers and suggestions with RAG	GenAI Integration Lead	Week 9	3
Frontend application	React.js app: CV upload, interview session, results dashboard	Frontend Developer	Week 10	3
Integrated MVP	Complete working system	All members	Week 10	4
Test and evaluation report	Test cases, results, model metrics, bug log	All members	Week 12	5
Deployed application	Production deployment with monitoring	Cloud / DevOps Engineer	Week 13	6
Final report, user guide and demo	Documentation and presentation	All members	Week 14	7

1.2.6 Alternative Baseline: Task Assignment Sprint Roadmap

Sprint length is assumed at 10 weeks in total and can be shifted to the course calendar without changing sprint goals. Detailed per-member sprint milestones and acceptance criteria are in Task Assignment and Rules, sections 2.2–2.7.

Sprint	Weeks	Theme	Exit gate (sprint review demo)
Sprint 1	1–2	Foundation and Contracts	Repository, CI skeleton, AWS baseline, DB schema v1, OpenAPI v0.1 frozen, UI wireframes, ML and GenAI proofs of concept.
Sprint 2	3–5	Core Components in Isolation	Each module works standalone: CV upload and parsing, benchmark and model selection, question generation, UI screens on mock API, staging environment.
Sprint 3	6–8	Integration and Intelligence	Full vertical slice on staging (UI → FastAPI → ML + GenAI → RDS → results); monitoring and alarms active; fallback modes working.
Sprint 4	9–10	Hardening, Testing and Delivery	Production deploy, load and security tests passed, final evaluation report, documentation, demo rehearsal, handover.

1.2.7 Task Dependencies

Dependency	Provider → Consumer	Needed by	Mitigation if late
OpenAPI contract v0.1	Sawsan → Anas, George, Belal	End Sprint 1	Technical Coordinator freezes contract; changes via versioned PR only.
DB schema v1 and migrations	Ahmed → Sawsan	End Sprint 1	Backend uses SQLite / Docker DB with the draft schema.
Mock ML and GenAI adapters	Sawsan → Anas	Mid Sprint 2	Frontend uses local JSON fixtures / mock server.
Staging environment and RDS	Mennatullah, Ahmed → All	End Sprint 2	Local docker-compose remains the integration baseline.
Scoring module (v1)	George → Sawsan	Start Sprint 3	Backend ships a lexical fallback scorer behind a feature flag.

Dependency	Provider → Consumer	Needed by	Mitigation if late
Question-generation service	Belal → Sawsan	Start Sprint 3	Backend serves a static fallback question bank.
Alarm and dashboard set	Mennatullah → All	Mid Sprint 3	Manual log-review checklist until alarms are live.

Phase-level dependencies follow the Gantt chart: integration (Phase 4) requires the backend, frontend, evaluation and GenAI modules from Phase 3; testing, deployment and documentation overlap in Weeks 10–14; and stretch goals start only after M5.

1.2.8 Resource Allocation

Team members and their roles are detailed in 1.3. Tools, infrastructure and services:

Resource	Purpose
AWS (EC2, S3, RDS, API Gateway, CloudWatch)	Hosting, file storage, database, API routing, monitoring
OpenAI API / Amazon Bedrock	LLM-based generation (usage budget To Be Confirmed)
GPU environment (Google Colab or AWS GPU instance)	Model experimentation and Whisper (if used)
GitHub and GitHub Actions	Version control, code review, CI/CD
Trello / Jira / GitHub Projects	Task tracking and sprint planning (tool not yet fixed)
Figma (or similar)	UI wireframes and mockups
Python, FastAPI, PyTorch / Hugging Face, Sentence Transformers, React.js	Core development stack

1.2.9 Testing, Documentation, Deployment and Presentation Milestones

- **Testing:** unit tests (backend target ≥ 80% line coverage), integration tests for the full flow, load tests on the deployed API, and model evaluation; milestone M6 (Week 12).
- **Deployment:** production deployment with a manual approval gate and CloudWatch monitoring; milestone M7 (Week 13). In the Task Assignment roadmap, production must be stable for 72 hours with rollback rehearsed (Sprint 4).
- **Documentation and presentation:** final report, user guide, demo preparation and presentation; milestone M8 (Week 14). Calendar dates and any course-mandated milestones are To Be Confirmed.

Definition of done (Task Assignment §3.3): style-guide compliance and peer review; unit and integration tests passing in CI; documentation, OpenAPI and changelog updated; inputs validated, no CV text in logs, least-privilege IAM; successful staging deployment with known rollback; acceptance criteria demonstrated.

1.3 Task Assignment and Roles

1.3.1 Team Roster and Component Ownership

#	Member	Role	Core stack	Owned component / repo area
1	Mennatullah Essam Mohamed Kamal Abdelfattah	Cloud / DevOps Engineer	AWS, GitHub Actions, Docker, IaC	Cloud and delivery: /infra, /.github, /docker
2	Sawsan Mahmoud Elgamal	Backend Developer	Python, FastAPI, Pydantic, SQLAlchemy, pytest	API and orchestration: /backend
3	George Ehab Sobhy Dos	Machine Learning Engineer	Whisper, Sentence Transformers, BERT / DistilBERT, PyTorch	Evaluation and speech-to-text: /ml
4	Ahmed Hamdy Ahmed	Database Engineer	PostgreSQL / MySQL, Alembic, AWS RDS	Data layer: /db, /backend/migrations
5	Anas Hany Shaker	Frontend Developer	React.js, JavaScript, REST clients, Jest / RTL	Presentation layer: /frontend
6	Belal Salah Hassan	GenAI Integration Lead	OpenAI API, Amazon Bedrock, prompt engineering, RAG	Generative AI: /genai

Names are shown in the form used in the SRS and Proposal where a full name is given; the Task Assignment uses shortened names. Team size is six members.

1.3.2 Responsibilities and Expected Outputs

Member	Responsibilities	Expected outputs
Mennatullah (Cloud / DevOps)	Provision and maintain AWS infrastructure and IAM (least privilege); own CI/CD, CloudWatch observability, cost governance, secrets, the S3 CV bucket and Bedrock access.	Infrastructure-as-code repository; Dockerfiles; GitHub Actions workflows; CloudWatch dashboards and alarms; deployment runbook and rollback procedure; per-sprint cost report.
Sawsan (Backend)	Own the FastAPI architecture; implement REST endpoints (authentication, CV upload, roles, sessions, questions, answers, results); integrate ML, GenAI and database modules; validation, JWT, rate limiting; publish the OpenAPI contract.	API skeleton and versioned endpoints; CV parsing service; ML / GenAI adapters with mocks; pytest suite (≥ 80% line coverage target); API documentation.
George (Machine Learning)	Build the answer-evaluation pipeline and calibrated 0–100 confidence score; derive strengths and weaknesses; create the human-rated benchmark; integrate Whisper (stretch); provide a lexical fallback scorer.	Benchmark dataset and labelling rubric; model comparison report (Sentence Transformers vs BERT vs DistilBERT); evaluation module; Whisper module with WER and real-time-factor results; model card and final evaluation report.
Ahmed (Database)	Design the relational schema (users, CVs, roles, sessions, questions, answers, evaluations, feedback, audit events); decide PostgreSQL vs MySQL; migrations; RDS configuration, backups and retention / deletion rules.	Schema v1 and Alembic migrations; seed data and indexing strategy; RDS configuration; retention rules; backup / restore test with Mennatullah.
Anas (Frontend)	Build the React.js application (authentication, CV upload, role selection, interview room, results dashboard, history); typed API client; responsive and accessible UI; frontend tests.	React.js application; UI wireframes and mockups; frontend unit and smoke tests.
Belal (GenAI)	Integrate OpenAI API / Bedrock behind a provider-agnostic interface; design and version prompts; build the RAG pipeline; implement guardrails (JSON-schema output, safety and bias filters, prompt-injection defences); control cost with caching and fallbacks.	Question-generation and ideal-answer functions; RAG knowledge base; prompt documentation; generation-quality evaluation.

Shared responsibilities: all members share testing, documentation and the final presentation. Pairing at integration points is encouraged (Backend with ML, Frontend with Backend, DevOps with Database).

1.3.3 Sprint Facilitation Roles

Role	Holder	Responsibility
Technical Coordinator	Sawsan Mahmoud (all sprints)	Owns the integration contract, arbitrates technical decisions, maintains the sprint board.
Release Manager	Mennatullah Essam (all sprints)	Tags releases, controls deploy windows, owns rollback.
Scrum Master (rotating)	S1 Sawsan | S2 Anas | S3 Ahmed | S4 George	Runs standups and retrospectives, tracks blockers, publishes sprint summary.
Scribe (rotating)	Rotates weekly	Records decisions in the decision log.

> *The Technical Coordinator and Release Manager assignments are Proposed in the Task Assignment ("confirm at kickoff") and are To Be Confirmed.*

1.3.4 RACI Matrix

R = Responsible, A = Accountable, C = Consulted, I = Informed (Task Assignment §2.8; each activity has exactly one A).

Activity	Menna	Sawsan	George	Ahmed	Anas	Belal
AWS infrastructure and IAM	A/R	C	I	C	I	C
CI/CD pipeline	A/R	R	C	C	R	C
FastAPI design and endpoints	I	A/R	C	C	C	C
Database schema and migrations	C	C	I	A/R	I	C
Answer-evaluation pipeline	I	C	A/R	I	I	C
Whisper speech-to-text	C	C	A/R	I	C	I
GenAI questions, prompts and RAG	C	C	C	C	I	A/R
React UI and UX	I	C	I	I	A/R	C
Security and privacy controls	A/R	R	I	R	R	R
Monitoring and cost control	A/R	C	C	C	I	R
Integration and end-to-end testing	R	A/R	R	R	R	R
Final report and demo	R	A/R	R	R	R	R

1.3.5 Governance, Conflict Resolution and Accountability

- **Conflict resolution:** (1) direct conversation within 24–48 h; (2) mediated discussion by the Technical Coordinator (technical) or Scrum Master (process); (3) team vote by simple majority, ties broken by the Technical Coordinator; (4) escalation to the course supervisor within 48 h of a failed step 3.
- **Missed deadlines:** early warning at least 24 h before a deadline; Level 1 recovery plan (1–2 days overdue); Level 2 re-plan by the Technical Coordinator (over 2 days or blocking a sprint gate); Level 3 escalation with contribution records shared with the supervisor.
- **Individual contribution** is evidenced by Git history, reviewed pull requests, closed tickets and sprint-review demos.

1.4 Risk Assessment and Mitigation Plan

The risk register below is taken from the Risk Assessment and Mitigation Plan. Likelihood (L) and Impact (I) use 1–5 scales and Score = L × I. The register classifies 4 risks as Critical (R03, R04, R05, R10; scores 15–16), 6 as High (R01, R02, R06, R08, R09, R11; scores 10–12) and 2 as Medium (R07, R12; score 9). Owners are as stated in the source.

1.4.1 Documented Risk Register

ID	Category	Risk	L	I	Score / priority	Mitigation strategy	Contingency plan	Owner
R01	Technical / Model	Whisper latency and noise sensitivity	4	3	12 High	• Benchmark model sizes (tiny / base / small) for WER vs. real-time factor in Sprint 1-2 • Pre-process audio: resample to 16 kHz mono, noise suppression, VAD trimming; cap clips at 90 s • Run transcription asynchronously with progress indicator; cache model in memory • Show transcript for user confirmation / edit before scoring	• Disable voice mode via feature flag; text answers remain the primary path (voice is a stretch goal) • Switch to smaller Whisper model or managed STT (Amazon Transcribe) if budget allows	George Ehab
R02	Technical / Model	Sentence-Transformer scoring inconsistency	3	4	12 High	• Build >= 200 human-rated benchmark with 2 raters; select model by Spearman / MAE, not intuition • Calibrate similarity to a 0-100 scale per role; combine with rubric coverage (key-point checklist) • Regression test set and determinism test (same input > same score within tolerance) • Robustness tests: very short, off-topic, repeated keywords, copy-pasted question	• Present score as a band (e.g., Developing / Solid / Strong) with explanation rather than a precise number • Fall back to lexical + rubric scorer; disclose limitations in the UI	George Ehab
R03	Technical / Model	LLM hallucination or toxic / biased interview questions	3	5	15 Critical	• Ground generation with RAG over curated corpus; instruct model to use only supplied CV and role text • Enforce JSON schema output; reject / regenerate invalid responses (max 2 retries) • Moderation + blocklist for protected-attribute questions; automated bias test set • Manual review of >= 50 outputs per sprint; thumbs-down feedback capture • UI disclaimer: AI suggestions may contain errors	• Auto-switch to curated static question bank for the affected role / session • Kill-switch flag disables generated ideal answers while keeping questions • Purge offending cached generations; add case to red-team set	Belal Salah
R04	Technical / Model	GenAI API rate limits and outages	4	4	16 Critical	• Provider-agnostic client with exponential back-off + jitter, request timeouts and circuit breaker • Cache generated questions per (role, CV hash); pre-generate question pool per role • Token budgets per session; queue and smooth burst traffic • Request quota increases early; confirm limits in Sprint 1	• Fail over to secondary provider (Bedrock <> OpenAI) via configuration flag • Fall back to curated static bank; UI shows degraded-mode banner • Demo mode with pre-recorded responses for live presentations	Belal Salah

ID	Category	Risk	L	I	Score / priority	Mitigation strategy	Contingency plan	Owner
R05	Cloud & Financial	Unexpected AWS costs	4	4	16 Critical	• AWS Budgets at 50 / 80 / 100% (actual) and 100% forecast with SNS to Discord / Slack • Cost-allocation tags on every resource; weekly cost review at sync • Right-size: small instances, single-AZ RDS for dev / staging, auto-stop non-prod outside working hours • Per-session token cap and daily LLM spend alarm; prefer cheaper models for drafts • IaC teardown scripts; avoid NAT gateway where possible (use VPC endpoints / public subnets with SG control)	• At 80% budget: freeze non-essential environments and switch GenAI to static bank / smaller model • At 100%: stop staging, keep only production demo stack, request credits from supervisor / AWS Educate	Mennatullah Essam
R06	Cloud & Financial	Quota exhaustion	3	4	12 High	• Audit quotas and request increases in Sprint 1-2 (Bedrock model access may take days) • Monitor AWS/Usage and Service Quotas metrics with alarm at 80% utilisation • Load test early (Sprint 3) to find limits before the demo window • Document free-tier expiry dates per account	• Redirect traffic to secondary region / provider where configured • Switch to cached / static content; scale down non-critical workloads to free quota	Mennatullah Essam
R07	Architecture & Integration	FastAPI-to-React latency	3	3	9 Medium	• Latency budget per hop; p95 targets in OpenAPI notes (non-AI <= 500 ms, evaluation <= 3 s) • Async endpoints; long tasks via background job + polling / Server-Sent Events • Warm model at startup; response compression; pagination; caching of static data • Frontend: optimistic UI, skeleton loaders, request timeouts with retry • Distributed tracing via correlation ID logged to CloudWatch	• Return partial results (score first, suggestions later) • Scale EC2 vertically for demo; disable non-essential enrichment calls	Sawsan Mahmoud
R08	Architecture & Integration	Database connection-pool bottlenecks under concurrent sessions	3	4	12 High	• Size pool = f(workers, max_connections); keep a safety margin; pool_pre_ping and recycle • Use RDS Proxy or PgBouncer if connection count approaches limit • Keep transactions short; no AI calls inside open DB transactions; statement and idle timeouts • Concurrency load test (>= 20 sessions) in Sprint 3-4; monitor DatabaseConnections • Indexes and query plans reviewed (EXPLAIN)	• Reduce workers / enable queueing and return HTTP 503 with Retry-After • Increase instance class temporarily; restart leaking workers via health check	Ahmed Hamdy
R09	Data & Security	CV privacy and data leaks during upload and storage	2	5	10 High	• S3: block public access, SSE-KMS encryption, presigned URLs with short expiry, lifecycle auto-delete • TLS everywhere; RDS encryption at rest; private subnets; least-privilege IAM • Upload validation: type / magic-bytes, size limit, AV scan, sanitised filenames • Never log CV text or PII; redact before sending to LLM where feasible (mask phone, email, address) • Explicit consent notice, retention limit and user-initiated deletion (DB + S3) • Use only synthetic / consented CVs in testing	• Incident response: revoke keys, disable upload endpoint, rotate credentials, snapshot logs • Notify affected users and supervisor; purge exposed objects; post-mortem within 48 h	Mennatullah Essam

ID	Category	Risk	L	I	Score / priority	Mitigation strategy	Contingency plan	Owner
R10	Data & Security	Prompt injection	4	4	16 Critical	• Separate system instructions from user content with delimiters; treat CV / answer text as data only • Strip / neutralise instruction-like patterns; length limits; PDF hidden-text extraction checks • Strict output schema validation; no tool or code execution with LLM output • Scoring performed by the NLP model, not the LLM, so injection cannot alter scores • Maintain red-team suite of >= 30 injection prompts run in CI before release	• Block / flag session; fall back to static bank for that user • Patch prompt and add test case; rotate keys if system prompt contained secrets (it must not)	Belal Salah
R11	Architecture & Integration	Cross-module integration delay	3	4	12 High	• Freeze OpenAPI v0.1 in Sprint 1; contract-first development; mocks for every module • Weekly integration check and dependency table review • Sprint 3 gate: full vertical slice on staging • Docker-compose environment for every member	• Ship the minimum viable slice using fallback modules (lexical scorer, static bank) • Technical Coordinator re-plans scope per accountability protocol	Sawsan Mahmoud
R12	Technical / Model	ML inference resource limits	3	3	9 Medium	• Profile memory per model; lazy-load Whisper only when voice is enabled • Use distilled models; optional quantisation / ONNX export • CloudWatch agent memory metric + alarm at 85%; container memory limits and restart policy • Separate ML container so it can be scaled independently	• Disable Whisper; use smaller embedding model • Move ML service to a larger instance for demo day only	George Ehab

Full risk descriptions, the heat map, the CloudWatch alarm catalogue (Risk Plan §4.2), severity levels and incident workflow (§4.3), and the GenAI and cost fallback ladder (§4.4) are in the Risk Assessment and Mitigation Plan and are not repeated here.

1.4.2 Risks Documented in the Project Proposal

Proposal risk	L	I	Proposal mitigation	Relation to register
Limited labeled data for evaluating answer scores	Medium	High	Team-reviewed synthetic Q&A; early human-rated benchmark; start with embedding similarity.	Partly covered by R02 (benchmark).
LLM API cost or latency exceeds expectations	Medium	Medium	Caching, token limits, usage caps, fallback model / provider.	Covered by R04, R05.
Inconsistent CV formats break parsing	High	Medium	Support PDF and DOCX first; test on varied CVs; editable parsed preview.	Not in register; retained from Proposal.
Privacy and security of uploaded CVs	Medium	High	Encrypted storage, access control, minimal retention, user consent.	Covered by R09.
Integration delays between modules	Medium	High	API contracts in Week 3; mock endpoints; continuous integration.	Covered by R11 (contract frozen in Sprint 1).
Score does not match human judgment	Medium	High	Early benchmark evaluation (Week 8); tune thresholds; combine similarity with key-concept coverage.	Covered by R02.
AWS cost overrun	Low	Medium	Free tier or credits; budget alerts; shut down idle resources.	Covered by R05.
Team member unavailability	Low	Medium	Cross-training through pairing, shared documentation, visible task board.	Not in register; see accountability levels in 1.3.5.

1.4.3 Additional Risks Identified in This Review (Recommendations)

The following risks arise from inconsistencies between the source documents. They are **Recommendation (review)** items; likelihood and impact are qualitative judgments, not source data, and owners are To Be Confirmed.

ID	Risk	Likelihood / impact	Suggested mitigation
RR-01	Schedule baseline ambiguity: 14 weeks / 7 sprints (Proposal) versus 10 weeks / 4 sprints (Task Assignment) versus 6 phases (SDD).	High / Medium	Adopt one baseline at kickoff, map it to calendar dates and update all documents.
RR-02	Conflicting acceptance thresholds (Appendix A, C-06 and Document 3) cause disputes at testing and evaluation.	High / Medium	Hold a single target-confirmation review before Sprint 3 and record outcomes in the decision log.
RR-03	Benchmark size inconsistent across sources (≥ 200 pairs vs 1,500 pairs with ≥ 3 raters), risking mis-sized labelling effort.	Medium / High	Confirm size and rater count in Sprint 1 against available time and budget.
RR-04	Voice scope undecided, so ML, frontend and test effort may be mis-planned.	Medium / Medium	Apply the Task Assignment gate: decide at end of Sprint 2 from benchmark results; keep text as the primary path.
RR-05	No calendar dates or course-mandated milestones are recorded, so deadlines cannot be tracked against the course.	High / High	Complete open decision A5 (dates) at kickoff.

1.5 Key Performance Indicators (KPIs)

1.5.1 KPI Governance and Status of Targets

The KPI Framework (v1.0) states that it is an approved baseline for capstone evaluation, but the source files contain no approval record, and other sources (Proposal, Task Assignment, Requirements extract) describe overlapping targets as proposed or to be confirmed. Targets in this section are therefore reported as **Stated in the KPI Framework** and must not be read as agreed requirements until the team confirms them (see 1.5.4).

Governance element	Definition (KPI Framework §1.2)
Ownership	Every KPI has one accountable owner (role) and one backup.
Measurement cadence	ML KPIs: per model release and weekly on a rolling production sample. Infrastructure KPIs: continuous (1-minute CloudWatch resolution). UX KPIs: weekly cohort review.
Status bands	Green = meets target; Amber = within 10% relative of target; Red = worse than the amber limit.
Escalation	Red for 2 consecutive windows → corrective-action ticket within 24 h; Red on a Critical KPI → immediate team review.
Release gate	A release candidate is promoted only if all Critical KPIs are Green and no more than 2 non-critical KPIs are Amber.

Table 1.5 – KPI lifecycle stages (KPI Framework §1.3)

Stage	Primary KPIs in focus
Development	WER, cosine calibration, Pearson / Spearman correlation, CV parsing accuracy
Integration and staging	API latency, error rate, RAG faithfulness, real-time factor
Pilot / demo	Completion rate, CSAT, availability, cost per session
Post-launch iteration	Drift indicators, hallucination rate, recalibration triggers

> *The KPI Framework places development in "Sprints 1–4" and integration in "Sprints 5–6", which does not match the 4-sprint Task Assignment roadmap or the 7-sprint Proposal schedule (Appendix A, C-01). The stage column below is mapped from the KPI Framework categories and is a review mapping.*

1.5.2 Project Delivery KPIs (Proposal)

KPI	Definition and measurement	Target	Stage	Basis
Milestone delivery	Milestones M1–M8 completed in the planned week (1.2.5).	On schedule; calendar dates To Be Confirmed	Weekly status; sprint review	Confirmed (Proposal)
End-to-end flow	Core flows work in a deployed environment (O1).	All core flows	M5, M7	Confirmed (Proposal)
Feedback coverage	Share of submitted answers receiving feedback (O4).	100%	Testing	Confirmed (Proposal)
Question relevance	Reviewer-rated relevance of sampled generated questions (O3).	≥ 80% (Proposal); ≥ 90% of 50 samples (Task Assignment)	Testing	Conflicting; To Be Confirmed
Documentation delivery	Final report, user guide and demo delivered (O6).	Week 14	M8	Confirmed (Proposal)

1.5.3 Technical, Operational and Usability KPIs (KPI Framework)

Responsible roles: ML = Machine Learning Engineer; GenAI = GenAI Integration Lead; BE = Backend; FE = Frontend; DB = Database; DevOps = Cloud / DevOps. Targets are as stated in the KPI Framework consolidated tracking matrix (39 KPIs); Critical flags and evaluation-set details are in KPI Framework §2–§4.

#	KPI	Category	Target	Measurement tool / source	Owner	Stage (mapped)
1	WER – clean English	ML / ASR	≤ 6%	jiwer on labeled clips	ML	Development
2	WER – accented English	ML / ASR	≤ 12%	jiwer	ML	Development
3	WER – noisy audio	ML / ASR	≤ 18%	jiwer, noise-augmented set	ML	Development
4	RTF (GPU)	ML / ASR	≤ 0.30	Timing harness	ML	Development
5	RTF (CPU fallback)	ML / ASR	≤ 0.80	Timing harness	ML / DevOps	Development
6	Pearson r vs. human	ML / Scoring	≥ 0.75	SciPy, benchmark set	ML	Development
7	Spearman ρ vs. human	ML / Scoring	≥ 0.72	SciPy	ML	Development
8	3-class macro-F1	ML / Scoring	≥ 0.78	scikit-learn	ML	Development
9	Expected Calibration Error	ML / Scoring	≤ 0.05	Custom ECE script	ML	Development
10	Keyword F1	ML / Scoring	≥ 0.82	scikit-learn	ML	Development
11	Scoring inference latency P95	ML / Scoring	≤ 250 ms GPU / 600 ms CPU	Application traces	ML / BE	Development
12	Context relevance	GenAI / RAG	≥ 0.80	RAGAS	GenAI	Integration / staging
13	Faithfulness / groundedness	GenAI / RAG	≥ 0.92	RAGAS / TruLens	GenAI	Integration / staging
14	Hallucination rate	GenAI / RAG	≤ 5%	LLM-judge + human audit	GenAI	Integration / staging
15	Time to first token P95	GenAI	≤ 1.5 s	Streaming logs	GenAI	Integration / staging
16	Latency per token	GenAI	≤ 40 ms mean	Streaming logs	GenAI	Integration / staging
17	Retrieval latency P95	GenAI / RAG	≤ 150 ms	Application traces	GenAI / DB	Integration / staging
18	LLM cost per session	GenAI / Cost	≤ US$0.08	Provider usage API / Bedrock metrics	GenAI / DevOps	Integration / staging
19	API P95 latency (light)	Backend	≤ 200 ms	CloudWatch / Prometheus	BE	Integration / staging
20	API P99 latency (light)	Backend	≤ 500 ms	CloudWatch	BE	Integration / staging
21	Full pipeline P95 latency	Backend	≤ 12 s	Locust load test	BE / ML	Integration / staging
22	Throughput	Backend	≥ 50 RPS	Locust	BE	Integration / staging
23	5xx error rate	Backend	≤ 0.5%	API Gateway metrics	BE / DevOps	Integration / staging
24	4xx error rate	Backend	≤ 3%	API Gateway metrics	BE	Integration / staging
25	Availability	Infra	≥ 99.5%	CloudWatch Synthetics	DevOps	Integration / staging
26	EC2 CPU utilization	Infra	≤ 65% avg (alarm 75/90)	CloudWatch	DevOps	Integration / staging
27	EC2 memory utilization	Infra	≤ 70% avg (alarm 80/90)	CloudWatch agent	DevOps	Integration / staging
28	RDS query time P95	Database	≤ 50 ms read / 100 ms write	RDS Performance Insights	DB	Pilot / demo
29	RDS connection pool usage	Database	≤ 70% of max	CloudWatch DatabaseConnections	DB / BE	Pilot / demo
30	S3 CV upload time	Storage	≤ 2 s P95	Client timing + S3 metrics	DevOps / FE	Integration / staging
31	S3 audio upload throughput	Storage	≥ 5 MB/s	Client timing	DevOps / FE	Integration / staging
32	Session completion rate	Product	≥ 75%	RDS event logs / dashboard	FE / BE	Pilot / demo

#	KPI	Category	Target	Measurement tool / source	Owner	Stage (mapped)
33	CSAT (feedback quality)	Product	≥ 80% positive	In-app survey (RDS)	FE / GenAI	Pilot / demo
34	Thumbs-up ratio	Product	≥ 0.80	In-app feedback (RDS)	FE	Pilot / demo
35	SUS score	UX	≥ 75	User study questionnaire	FE	Pilot / demo
36	CV parsing accuracy	Functional	≥ 92% field-level	Labeled CV set (n = 200)	ML / BE	Pilot / demo
37	CV extraction latency P95	Functional	≤ 5 s	Application traces	BE	Pilot / demo
38	Test coverage	Quality	≥ 80%	pytest-cov / CI	BE	Pilot / demo
39	Monthly AWS cost	Cost	≤ US$150	AWS Budgets / Cost Explorer	DevOps	Pilot / demo

1.5.4 Conflicting and Unconfirmed Targets

Metric	Proposal	Task Assignment	KPI Framework	Requirements extract
Score vs human correlation	Spearman ≥ 0.6 (e.g.; set during evaluation)	ρ ≥ 0.60, to be confirmed after Sprint 2	Pearson ≥ 0.75; Spearman ≥ 0.72 (Critical)	No threshold
Evaluation latency	Average under about 10 s	p95 ≤ 3 s	p95 ≤ 3.0 s (excl. ASR)	95% within 5 s
Lightweight API latency	—	p95 non-AI ≤ 500 ms at 20 concurrent sessions	p95 ≤ 200 ms; p99 ≤ 500 ms	95% within 2 s
Question relevance	≥ 80% sampled	≥ 90% of 50 samples	Mean quality ≥ 4.0 / 5	—
Concurrency / load	—	20 concurrent sessions	≥ 50 RPS lightweight	25 concurrent users
Availability	—	—	≥ 99.5% monthly	≥ 95% in evaluation period
Benchmark size	Small human-rated set	≥ 200 pairs (Risk R02: 2 raters)	1,500 pairs, ≥ 3 raters	—
Budget	To be confirmed	Approved budget (value not stated)	US$150 per month	—

No conflict is resolved here because no source establishes which value is authoritative. **Recommendation (review):** confirm one value per metric before Sprint 3 and record it in the decision log; until then, treat all values in this section as Proposed.

Appendix A – Source Conflicts and Open Decisions (Planning)

ID	Topic	Conflicting statements	Status
C-01	Schedule	Proposal: 14 weeks, 7 two-week sprints. Task Assignment: 10 weeks, 4 sprints. SDD: 6 phases (about 7–9 weeks). KPI Framework refers to Sprints 5–6.	Unresolved; Proposal used as plan baseline
C-02	Calendar dates	No source gives calendar dates (Task Assignment A5, due at kickoff).	To Be Confirmed
C-03	Voice input	Stretch goal (Proposal, Task Assignment, gated by Sprint 2 benchmark) versus in-scope optional feature (SRS, Requirements extract).	Unresolved (decision A4)
C-04	Database engine	PostgreSQL / MySQL (Proposal, SDD, Requirements); PostgreSQL with pgvector (SRS DC-2); Task Assignment recommends PostgreSQL, decision A1 due end of Sprint 1.	Open; PostgreSQL recommended
C-05	Vector store for RAG	pgvector vs ChromaDB (Task Assignment A3); pgvector / FAISS (Feedback Specification); pgvector in PostgreSQL (SRS).	Open (decision A3)
C-06	Acceptance targets	See 1.5.4.	Unresolved
C-07	GenAI provider	OpenAI API or Amazon Bedrock, failover order undecided (Task Assignment A2).	Open
C-08	Orchestration framework	Not fixed (Proposal assumption; for example LangChain).	Open
C-09	Objective numbering	Proposal O1–O6 versus Task Assignment O1–O5 with different content.	Proposal numbering used here